Vanessa's summer plan

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1 How I view this project

Parts: population, parameter estimation, lensing
i. LVK data

Using parameters(usually 15) converted from string data (waveform) to plot density distribution vs each parameter, e.g. primary mass, X eff

ii. Samplers

Using Baye's rule to calculate posterior which is an inverse problem and use stochastic samplers: Markov-chain Monte Carlo

iii. Model selection

Use Baye's factor to compare fully marginalized likelihood functions Z, called evidence. There can be different priors and can be two models with different parameters or even different numbers of parameters. Tell which prior or model to select

2 What I need to do

i. Background knowledge (reading):

1. Hierarchical Bayesian analysis ***
2. Current LVK population results
3. Formation channel (Optional)
4. Gaussian Mixture
5. Help from the lensing people

ii. Technical/ coding

1. reproduce LVK results with FIGARO
2. FIGARO with lensing data?

3 Goal

i. primary mass distribution
ii. spin distribution